

REPUBLIC OF SOUTH AFRICA

PATENTS ACT, 1978

APPLICATION FOR A PATENT AND ACKNOWLEDGEMENT OF RECEIPT

[Section 30 (1)-Regulation 22]

The granting of a patent is hereby requested by the undermentioned applicant on the basis of the present application.

Official Application No.		Applicant's or Agent's Reference
21	01	6711LPS
2026/04859		

71	Full Name(s) of Applicant(s)
VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India	

54	Title of invention
A PRIORITY-BASED TRAFFIC MANAGEMENT SYSTEM FOR AMBULANCES USING AI-DETECTION AND IOT-BASED LORA COMMUNICATION	

The applicant claims priority as set out on the accompanying Form P.2. The earliest priority claimed is		
COUNTRY:	NUMBER:	DATE:

This application is for a patent of addition to patent application No.		
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21	01	
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This application is a fresh application in terms of section 37 and based on Application No.		
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21	01	
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This application is accompanied by:		
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X	1.	A single copy of a complete specification of 10 pages.
X	2.	Drawings of 2 sheet(s).
X	3.	Publication particulars and abstract(Form P8)
X	4.	A copy of a figure of the drawing (if any) for the abstract
X	5.	Assignment of invention
	6.	Certified priority document(s)
	7.	Translation(s) of the priority document(s)
	8.	Assignment of priority rights
	9.	A copy of the Form P.2 and the specification of S.A Patent Application (if applicable).
X	10.	A declaration and power of attorney on Form P3
X	11.	Statement on the use of indigenous Biological Resource, Genetic Resource, Traditional Knowledge or Use on Form P26

74	Address of Service:
Wolmarans and Susan Inc. 337 Surrey Avenue, Randburg, Gauteng, 2194 SOUTH AFRICA	

Dated this 6th day of May 2026

Digitally signed by : Leon Pierre Susan

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Signature of Applicant(s)

This is returned to the applicant's
address for service as proof of lodging.

RECEIVED
Official Date Stamp
..... Registrar of Patents

REPUBLIC OF SOUTH AFRICA

REGISTER OF PATENTS

PATENTS ACT, 1978

Official application No.		Lodging date: Provisional		Acceptance date	
21	01	2026/04859		22	
International classification		Lodging date: Complete		Granted date	
51	G08G	23	2026/05/06		
71 Full name(s) of applicant(s)/Patentee(s):					
VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India					
71 Applicant substituted:				Date registered	
71 Assignee(s):				Date registered	
72 Full name(s) of inventor(s):					
ZADE, Anuja AVALLE, Ajay Mahadev MITKARI, Oshan Amol RATHOD, Chaitanya Bhaurao BHUJBAL, Ketan Kaluram THAKUR, Dhruv Manoj					
Priority claimed:		Country	Number	Date	
54 Title of invention					
A PRIORITY-BASED TRAFFIC MANAGEMENT SYSTEM FOR AMBULANCES USING AI-DETECTION AND IOT-BASED LORA COMMUNICATION					
Address of applicant(s)/patentee(s):					
666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037 INDIA					
74 Address for service					
Wolmarans and Susan Inc. 337 Surrey Avenue, Randburg, Gauteng, 2194 SOUTH AFRICA Reference No. 6711LPS					
61 Patent of addition No.			Date of any change		
Fresh application based on.			Date of any change		

REPUBLIC OF SOUTH AFRICA
PATENTS ACT, 1978
COMPLETE SPECIFICATION
[Section 30(1) – Regulation 28]

OFFICIAL APPLICATION NO.

21	01	2026/04859
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LODGING DATE

22	2026/05/06
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INTERNATIONAL CLASSIFICATION

51	G08G
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FULL NAME(S) OF APPLICANT(S)

71	VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India
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FULL NAME(S) OF INVENTORS(S)

72	1. ZADE, Anuja 2. AVALE, Ajay Mahadev 3. MITKARI, Oshan Amol 4. RATHOD, Chaitanya Bhaurao 5. BHUJBAL, Ketan Kaluram 6. THAKUR, Dhruv Manoj
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TITLE OF INVENTION

54	A PRIORITY-BASED TRAFFIC MANAGEMENT SYSTEM FOR AMBULANCES USING AI-DETECTION AND IOT-BASED LORA COMMUNICATION
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REPUBLIC OF SOUTH AFRICA
PATENTS ACT, 1978
PUBLICATION PARTICULARS AND ABSTRACT
[Section 32(3)(a) – Regulation 2291(g) AND 31]

OFFICIAL APPLICATION NO.			LODGING DATE		ACCEPTANCE DATE	
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INTERNATIONAL CLASSIFICATION		NOT FOR PUBLICATION
51	G08G	CLASSIFIED BY: Wolmarans and Susan Inc.

FULL NAME(S) OF APPLICANT(S)	
71	VISHWAKARMA INSTITUTE OF TECHNOLOGY 666, UPPER MARKET RD, UPPER INDIRA NAGAR, BIBWEWADI, PUNE, MAHARASHTRA, 411037, India

FULL NAME(S) OF INVENTORS(S)	
72	1. ZADE, Anuja 2. AVALE, Ajay Mahadev 3. MITKARI, Oshan Amol 4. RATHOD, Chaitanya Bhaurao 5. BHUJBAL, Ketan Kaluram 6. THAKUR, Dhruv Manoj

EARLIEST PRIORITY CLAIMED		DATE	
COUNTRY	NUMBER		
33	31	32	

TITLE OF INVENTION	
54	A PRIORITY-BASED TRAFFIC MANAGEMENT SYSTEM FOR AMBULANCES USING AI-DETECTION AND IOT-BASED LORA COMMUNICATION

57	A priority-based traffic-management system for ambulances is disclosed, wherein an ambulance unit uses an ai-enabled camera and a microphone to detect an approaching ambulance through visual recognition and siren-pattern analysis. A processing unit generates an emergency-priority signal and transmits the signal through a lora communication module to a traffic control unit. The traffic control unit preempts traffic signals so that the ambulance direction is turned green and other directions are turned red, thereby providing a clear path through the intersection. A lora-based manual override mechanism allows the ambulance driver to trigger signal priority when ai-based detection is unreliable due to poor visibility, obstruction, weather, or noise. The system provides a hybrid, fault-tolerant, internet-independent, and scalable solution for ambulance traffic-signal prioritization.
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